

Reporting Module — Annex

Field	Value
Companion to	PRD-reporting-module-v1
Owner	Mohamed Moanes (Functional Lead / acting Product Manager)
Date	2026-09-08
Source brief	SB-20260908-reporting-module-v1 (contract v2, Status: Ready for PRD, Content mode: PM-provided, confirmed by the Product Manager 2026-09-08)
Document control	v1, 2026-09-08. First version
What this is	Everything a reader looks up rather than acts on. Nothing here is optional content that was cut — it is reference material relocated so the main document can hold its page target

Who should read what

If you are	Read
UI/UX Design	A, B, C — the field dictionary, the screen states, the failure states
Engineering	E, F, G, H — the architecture reasoning, the requirement rationale, how each check runs, the dependencies
Product Owner	All of it, eventually. K is the audit trail
Business or leadership	Nothing here is aimed at you. J and K if you want to see what was rejected and what nobody has confirmed

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A. Detailed field specifications

Marked **Required**, **Optional** or **System-managed**. Every quantity follows Requirement 25 — stored in its base unit, scaled at display, label derived from the value.

The report definition

Field	Required?	Notes
Name	Required	Unique within its type and client account. Where two share a name, neither is ever shown without something distinguishing them
Description	Optional	Free text
Type	Required	System, private, or shared. Set at creation; changing it is a privileged action
Source dashboard	Required	Exactly one saved dashboard, default or customer-built. Named in the produced file (Requirement 2)
Source last-changed time	System-managed	Recorded on each run so a reader can see why two periods look different (Requirement 22)
Coverage kind	Required	Sites, devices, or portfolio. Portfolio offered only where the source permits it (Requirement 8)
Covered scopes	Required	One or more sites or devices, resolved against the polymorphic scope hierarchy. Constrained by the source dashboard and by the authority's access
Period	Required	The window the report covers. Defaults from the source dashboard's saved default and may be overridden
Output format	Required	PDF or Excel
PDF content choice	Required when format is PDF	Diagrams and tables, diagrams only, or tables only
Declared authority	System-managed	Whose access the dispatch resolves against (Requirement 19). Named in the produced file. Its assignment rule is the blocking question
Active	Required	The switch. Off by default on creation
Created by / created at	System-managed	
Last edited by / at	System-managed	

The schedule

Field	Required?	Notes
Kind	Required	Repeating, or one-time
Start date	Required for repeating	
Time of day	Required for repeating	

Field	Required?	Notes
Time zone	Required for repeating	Declared and shown on screen (Requirement 10). Never implicit
Repeat interval	Required for repeating	[NEEDS INPUT: which intervals – daily, weekly, monthly, or a stated set]. The Product Manager described "periodically" without naming the set
Next run at	System-managed	
Last run at	System-managed	

The recipient list

Field	Required?	Notes
Address	Required	At least one. A recipient need not be a platform user (Requirement 13)
Inside or outside the organization	System-managed	Derived from the address. Gates the outside-organization privilege
Added by / at	System-managed	

No recipient-side permission check exists, by design, and it is named as a non-goal in the main document. There is nobody to check.

A run record

Field	Required?	Notes
Report	System-managed	
Started at / finished at	System-managed	
Coverage and period as run	System-managed	Recorded as resolved, not as configured — they can differ if access narrowed
Authority as resolved	System-managed	
File reference	System-managed	Through the storage abstraction. Subject to the retention period
Recipients as sent	System-managed	Recorded before sending (Requirement 16)
Outcome	System-managed	Succeeded, failed with reason, or suppressed with reason
Attempt number	System-managed	Bounded retries per Requirement 17

B. Screen state matrix

Screen	Empty	Loading	Forbidden	Error	Partial data
Reporting setup page	A tab with no reports says so and offers the way to make one, where the privilege allows. The system tab is never empty	The tab shape and the badges resolve first, the list after	A person without the see-system privilege does not get the system tab at all — hidden, not empty	The list fails to load with a retry, and the tabs stay usable	Some reports show a "source unavailable" marker rather than the whole list failing

Screen	Empty	Loading	Forbidden	Error	Partial data
Report editor	A new report shows the source picker first, because every other field depends on it	Coverage options load after the source is chosen and are visibly pending	Controls the person lacks are absent, with no gap left in the layout	A save failure keeps every entered value. Nothing is silently discarded	If the coverage list is partly resolvable, show what resolved and name what did not
Run history	A report that has never run says so plainly, not as an error	Newest first, paged	Absent for a person without the read-history privilege	A history that will not load does not block the report itself	A run mid-flight shows as running, not as failed
Way in from a dashboard	N/A — reached from a dashboard that exists	Inline	Hidden where the create privilege is absent	Falls back to the setup page with the source pre-chosen	N/A
The produced PDF	A period with no readings produces a file that says there were no readings — never an empty page and never a silent success	N/A	N/A	A panel that cannot render leaves a stated placeholder, not a blank	Rendered panels appear; unrendered ones are named
The produced Excel	Same rule — a sheet stating no readings, not an empty sheet	N/A	N/A	A panel that cannot resolve gets a sheet naming the problem	As above
The delivery email	N/A	N/A	N/A	A failure never sends a partial email. Nothing goes out, and the run records why	N/A

One rule across all seven: nothing in this module ever reports success for an artefact that is not there. Silent non-delivery is the failure mode this whole matrix exists to prevent.

C. Failure states in full — these are requirements, not notes

#	Situation	Required behavior
1	The source dashboard is deleted while a schedule is live	Deletion names every dependent report first, and how many (Requirement 21). If deleted anyway, those reports go inactive with the reason recorded. No file is sent
2	The source dashboard is edited	The next run reflects the edit. The file records the source's last-changed time (Requirement 22). No notification — an edit is normal
3	The declared authority loses access to a covered site	The next run does not send. The report goes inactive with the reason recorded (Requirement 20). The file is not trimmed to what remains visible — a report that silently shrinks is worse than one that stops
4	The declared authority's account is	Same as 3

#	Situation	Required behavior
	switched off	
5	The generated file exceeds the attachment ceiling	The run fails before sending, with a stated reason. The file is retained so somebody can retrieve it. Never handed to a mail provider to drop
6	A run fails — renderer, store or mail	Recorded with its reason, retried a bounded number of times, then the report goes inactive and its creator is told (Requirement 17)
7	Many schedules fire in the same minute	Runs queue rather than being dropped. Ordering is not guaranteed; delivery is. [NEEDS INPUT: how many concurrent runs must be supported]
8	A recipient address bounces	The run is recorded as delivered-with-failures, naming which addresses failed. Delivery to the others is not rolled back — email cannot be un-sent
9	The period contains no readings	A file is still produced, and it says there were no readings. See the state matrix
10	A one-time report is requested twice in quick succession	Two runs, two files, two deliveries. This is not deduplicated — the person asked twice
11	The report is switched off and on across a missed run	No backfill (Requirement 12). The next run is the next scheduled one

D. Quality bars in full

Quality bar	Requirement
Delivery reliability	Every scheduled run either delivers or is recorded as failed with a reason. Silent non-delivery is a defect
Attachment ceiling	A file too large to email fails the run visibly, before sending. Ceiling: [NEEDS INPUT]. Architect-proposed — confirm
Separation	Structural, inherited from the Admin Module — a client identifier on every top-level record and a per-client sign-in space. Never a query filter
Mail route	Through the platform's mail abstraction only. Verifiable in code as well as in behavior (Check 9)
Authority at dispatch	Resolved at the moment of the run, never cached from creation time
Retention	[NEEDS INPUT: how long is a generated file kept, and does any regulation apply]
Generation time	[NEEDS INPUT] — nobody specified how long a report may take to build
Concurrency	[NEEDS INPUT] — how many runs may fire in the same minute
Uptime	[NEEDS INPUT]
Accessibility	[NEEDS INPUT] — no requirement was given, and none is invented
Volume	[NEEDS INPUT] — sites, devices, users, active schedules. The most consequential gap in this table , because the scheduler cannot be sized without it

D2. The Product Manager's own words, in full

Quoted verbatim in Idea Brief SB-20260908-reporting-module, dated 2026-09-08, and reproduced here because they are the only record of the original phrasing and the routing in Data & Integration was matched against them rather than against a paraphrase.

"It reads all kind of renewable energy Sites Readings & KPIs or any sites that have energy and power readings in forms of Widgets and Diagrams and tables"

"they are all exported from the system default dashboards and custom dashboards ... Then the user is able to select the applied sites, devices or even on the portfolio according to the selected dashboard in the dropdown list scope ... Then the user will be able to set the report output whatever it will be PDF or Excel Sheet, if PDF whatever it will contain diagrams and tables, diagrams only, or tables only ... Finally the user will put all the mail/s that want the report to be sent to, so that the report will be sent automatically in the scheduled date and timing"

"the user can create from his default / customizable dashboards, scheduled reports that will be sent periodically to all the mentioned email in the reporting setup page"

On the problem today:

"can only create it manually and can only send it manually not scheduled"

On the driver:

"they are doing everything manually" · "we want them to use our reporting module instead"

The quotation is in ordinary words throughout, with no engineering vocabulary in it. That is what the mapping table in INTAKE_GUIDE.md needs, and it is why the routing in Data & Integration could be done at all. Worth stating: this brief got the single most important thing right.

E. Sand Fit — the reasoning

Owning package or bounded context — candidates

All candidates. All must be confirmed with architecture. The digest confirms seven package names across four layers as of 2026-08-02: scoping, entityreg, platform (Foundations) → sysmodel, datascope, sysmon (System) → dataflow (Data Processing) → energyres (Domain). **There is no reporting package among them.**

Concern	Candidate home	Reasoning
The report definition — what a report is, its privileges, its lifecycle	Wherever a saved dashboard view lives. PRD-customizable-dashboards-v2 left that unresolved as <i>"where a saved view lives"</i>	A report definition and a saved view are the same kind of artefact: a named, client-scoped, privilege-gated composition over measures. Two homes for one shape would be a mistake, and answering the question once serves both documents
Coverage resolution — sites, devices, portfolio	scoping, which the digest names as the owner of the polymorphic Scope base, Measure and Quantity	Coverage is not a new concept. It is scope selection, and scope authorization already governs it
Value resolution	dataflow — DataAcquisitionService, the ReadingFrame family — plus the measure pipeline	The report reads what the dashboard reads. Requirement 7 exists to keep it to one path
Rendering to PDF and Excel	Unknown. Nothing in the digest names a renderer	Named as an open question rather than placed. A guess here would be invention
Time-triggered dispatch	Unknown, and this is the significant gap. platform owns app lifecycle, tenancy and entitlements, which is the closest thing to infrastructure among the Foundations packages	But a low-layer scheduler that reads measures would depend upward, which is illegal — see Dependency direction below
Delivery	The mail abstraction, reached through the adapter pattern. PRD-notification-	This document adopts that language deliberately, so two modules do not invent

Concern	Candidate home	Reasoning
	center-v5-ANNEX names <i>"the platform's mail abstraction. Never a direct connection to a mail provider"</i>	two mail routes
The setup page and the Generator measure catalogue	app:sandgen	Application vocabulary. Sand Distribution will need a different catalogue over identical machinery

Placements rejected, and why

Rejected	Why
The whole module inside `app:sandgen`	01_company-and-product-overview.md lists Reporting as a centralized backend engine shared across all frontends ; Sand Distribution's D0.5 slot needs the same capability; and PRD-customizable-dashboards-v2 already names the Report Creator as the next consumer of the platform composition model. Building it in the app guarantees a second implementation when Sand Distribution starts
A new reporting microservice	That is a deployment decision, not a code-architecture one. Restated as a bounded context
A general "delivery framework" any module can plug into	The upstream Project floated this and the Product Manager did not choose it. It is also the Generative claim this document declines to make. Two registered renderers and one delivery route is the minimum that satisfies the driver
Putting everything in `platform` to keep the imports legal	The tempting wrong answer. Placement is bounded-context semantics, never dependency legality — see below
A second calculation path optimized for reports	Requirement 7 forbids it. A report whose numbers differ from the dashboard it came from is worse than no report

Reuse before invent

Mechanism	Used here?	How
Adapter-based infrastructure abstraction	Yes, twice	The mail route and object storage, both through kernel abstractions. Neither coupled to a concrete provider. Canonical decision candidate: ADR-006
Structured flexibility — JSON parameterization	Yes	A report definition is configuration. New report shapes are data, not migrations. Whether this holds depends on the where-it-lives question
Polymorphic domain hierarchy	Yes	Sites, devices and portfolio <i>are</i> Scope. New coverage kinds would register rather than be coded
Registry-driven extension	Yes	Output formats are registered renderers (Requirement 4), so a third format costs a registration
Automatic CRUD resource generation	Yes, both halves	The report definition as a <code>crud_resource</code> ; dispatch as a <code>service_resource</code> . They are mutually exclusive per entity, so these are deliberately two entities
Schema-driven dynamic views	Yes	Rendering reads the same document-canvas-element chain the screen renders from. The PDF and the screen then agree by construction rather than by a second layout

Mechanism	Used here?	How
		implementation

What this leaves as genuinely new: four things. The report definition and its privileges, two renderers, a timer, and delivery. **The timer is the only one with no mechanism to extend**, which is why it is a blocking question rather than a task.

Extension without modification

Report types, coverage kinds and output formats all arrive by registration. Nothing in this module should require a kernel or core edit — **except possibly the scheduler**, and if it does, that is a signal about the boundary rather than a task to absorb quietly. Flag it loudly if the answer to the scheduler question is "build one in core."

Dependency direction

Downward only: Foundations → System → Data Processing → Domain → Apps.

The tension, stated plainly: **dispatch and delivery feel like Foundations concerns; reading measures is Data Processing.** A single reporting component low in the stack would have to import upward from dataflow, which is illegal. The likely resolution is a split — a generic scheduling and delivery primitive low down, with no knowledge of reports, and the report definition and orchestration above dataflow. **That is architecture's call.** Recording it as a tension rather than resolving it downward is deliberate: relocating a domain concern to a lower layer because the imports are more convenient there is the anti-pattern, not the fix.

Architecture decisions plausibly touched — every one a candidate for architecture to confirm

Named from 08_sand-fit-architecture-digest.md, **generated 2026-08-02. Never cited from memory, and never treated as settled.**

Decision	Why it is relevant here	Confidence
ADR-006 — adapters and the generic sub-domain boundary	The mail route and object storage. The canonical anti-corruption-layer decision	Verified present in the digest. Must be confirmed with architecture
ADR-022 — polymorphic domain events	Run history, and "who scheduled what"	Verified present. Must be confirmed
ADR-023 — cookie-based sign-in through the proxy	Everything here is behind sign-in, which sets the interface rung as the verification floor	Verified present. Must be confirmed
ADR-024 — hierarchical scope authorization	Coverage, and the declared-authority question	Verified present. Must be confirmed
ADR-029 — subject-display read-side element	The panels a report renders. The digest places this in the frontend	Verified present. Must be confirmed
ADR-035 — measure pipeline two-level resolution	KPI and measure resolution for the report's period	Verified present. Must be confirmed
The canvas-element-substrate decision	The composition chain a report renders from	One of two files both numbered ADR-031. Cite by filename, never by number
The measure-typing-and-aggregation decision	How values combine over the report's period	The other ADR-031. Same rule. This one is load-bearing here
ADR-033 — billing domain architecture	Named only to mark the boundary: billing documents are a non-goal	Verified present. Not used

Two things about the digest itself, found while writing this section

1. **The language-mapping table in `INTAKE\_GUIDE.md` has no row for a timer.** It maps "notify when X happens" to the event bus. Nothing maps "every Monday at seven." For a platform whose roadmap contains a Reporting slot, that is a gap in the routing aid, not just in this document. 2. **`packages/analytics` still appears in the digest's verified file**

locations and in none of its four package layers. PRD-customizable-dashboards-v2 raised this on 2026-08-10 and it is unresolved a month later. It is a likely home for part of this work.

F. Requirement rationale

Only where the reason is not obvious from the requirement.

#	Why it is worded that way
Requirement 2	<i>Exactly one</i> source dashboard. Multiple sources would make coverage, period and authority ambiguous, and nobody asked for it
Requirement 4	Formats as registered renderers rather than a branch. The Product Manager named two; the research found a third (comma-separated) that works only for tables. Registration means that judgement can be revisited without a rewrite
Requirement 6	The most useful thing the brief's research turned up: PDF and Excel are not symmetric . A PDF reproduces what is displayed; a spreadsheet puts each panel on a sheet; a chart-only dashboard produces nothing in a tabular export. Carrying the values behind each chart resolves it without asking the user to understand the asymmetry. Architect-proposed — confirm , and it is in Open Questions
Requirement 7	One calculation path. This mirrors Requirement 13 of PRD-customizable-dashboards-v2 rather than restating it — that PRD forbids a second path for formulas; this one forbids it for reports
Requirement 10	A time zone is not a detail. "Sent on the right day" is undefined without one, and a report that arrives on Sunday night for a Monday-morning meeting is a defect somebody will report as a bug
Requirement 12	No backfill . Switching a report back on after two weeks off should not send fourteen files. Nobody specified this; it is the less surprising behavior
Requirement 16	Record before send , because email cannot be recalled. PRD-notification-center-v5 reached the same conclusion about notification emails, and the reasoning transfers exactly
Requirement 18	Written against privileges , not roles, per the standing guidance in KNOWN_SOURCE_AMBIGUITIES.md A7: once custom roles ship, neither of Sand's two role lists is authoritative and the customer's own configuration is. A requirement naming a role would break the first time a client renames one
Requirement 19	The whole shared-report problem in one line. A dashboard resolves access when opened ; a file has no opening. So the authority must be decided at the run and named in the file , or a reader cannot tell whose portfolio they are holding
Requirement 20	Stopping is deliberately preferred to trimming. A report that silently covers fewer sites than last month looks like a business change rather than a permissions change
Requirement 21	Aligned with Requirement 15 of PRD-customizable-dashboards-v2, which refuses to delete a formula while dependents exist and degrades dependents to a placeholder if forced. Same shape, same reasoning, one more dependent type
Requirement 27	The G1.3 reporting capability shipped and works. Additive-first means it is untouched, not migrated

G. How each check is performed

#	Method
Check 1	Create a definition through the real endpoint with a signed-in session; read it back; assert every field persisted including the schedule and recipients

#	Method
Check 2	Request coverage outside the source dashboard's permitted scope, and separately request a site by identifier that the authority was never granted. Assert refusal at the fetch , and assert no partial file was produced
Check 3	Generate three PDFs, one per content choice. Assert programmatically that a diagrams-only file contains no table content and a tables-only file contains no chart content
Check 4	Generate an Excel file from a dashboard with mixed panels and from one with only diagrams. Assert one sheet per panel in both cases, and series values present for chart panels
Check 5	Command line with a controlled clock. Advance across a scheduled boundary and assert exactly one dispatch. Advance across two boundaries and assert two. Repeat with the report's time zone set to two different zones and assert the fire time differs accordingly
Check 6	Switch off, advance the clock past a boundary, assert nothing sent. Switch on, assert the next run is the next boundary and no missed run is emitted
Check 7	One-time request; assert one file, one run record, one dispatch
Check 8	Deliver to a captured mail sink including an address with no platform account. Assert the file arrives and the run record names every recipient
Check 9	Static inspection plus a run with the mail adapter unconfigured. Assert the failure comes from the abstraction, which proves nothing bypasses it
Check 10	Interface rung for the refusal, browser rung for the absence. Assert the control is not in the page , not merely disabled, and that the layout has no gap
Check 11	Run a shared report; assert the resolved authority in the run record and that the produced file names it
Check 12	Narrow the authority's access to exclude a covered site, advance to the next boundary, assert no dispatch, the report inactive, and a reason recorded
Check 13	Attempt to delete a source dashboard; assert the dependent reports are named with a count. Force the deletion; assert the reports go inactive with a reason and that no file is produced on the next boundary
Check 14	From a second client account, attempt to read the definition, the file reference and the run history by identifier, by list endpoint and by any export route. Assert every one refused
Check 15	Build a source dashboard scoped to one site; assert portfolio coverage is not offered, and that requesting it by identifier is refused
Check 16	Resolve the source dashboard's values through the interface for a coverage and period, generate a report for the same, and compare the numbers exactly
Check 17	Assert the run record exists with its recipients before the mail sink receives anything — ordering is the point of the check, not the presence of the record
Check 18	Retrieve the stored file through the storage abstraction after the run. Retention duration is not checked, because the duration is unspecified
Check 19	Force a renderer failure and separately a mail failure. Assert the reason recorded, the bounded retry count observed, the report inactive afterwards, and the creator informed
Check 20	Assert values in the produced PDF and Excel are scaled from the base unit and that the label matches the magnitude — a value stored in watt-hours displaying as megawatt-hours where appropriate
Check 21	Browser rung. Exercise every list control and assert the counter and badges agree with the underlying counts

#	Method
Check 22	Browser rung. Walk the pre-existing manual report path and assert unchanged behavior
Check 23	Operator-confirm. A person opens a PDF beside the dashboard it came from and agrees it is recognizably the same thing — ordering, grouping, legibility, nothing cut off. Gates the release close , never the dispatch of work, and stands in for no lower rung

H. Dependencies in full

Depends on	State	Source	Consequence if not there
Admin Module — multi-user accounts and four predefined roles	In user acceptance testing per the Product Manager, 2026-09-08. 09_shipped-state.md (2026-08-09) still records it in progress	Brief + PM; shipped-state re-checked	Private and shared report types are meaningless. Only system reports could exist
Diagram Builder and Widget Builder / customizable dashboards	In user acceptance testing per the Product Manager, 2026-09-08. 09_shipped-state.md records it in progress	Brief + PM; shipped-state re-checked; PRD-customizable-dashboards-v2	No custom sources. Only system default dashboards could be reported on
A working email route	In the pipeline with no date , per PRD-notification-center-v5. A mail <i>abstraction</i> is named; a delivered route is not	This document's own check — not in the brief	Nothing delivers. The single largest dependency risk, and the reason Stage 1 avoids it
SandGen default dashboards	Shipped 2026-08	09_shipped-state.md	The system report type has no sources
Formula Builder	Postponed	PM, 2026-09-08	Nothing — it is a non-goal. Listed so nobody assumes formula-sourced reports are pending rather than excluded
A time-triggered scheduler	Unknown. Named nowhere in the digest	This document's own check	The core mechanism has no home. Blocking question
A PDF / spreadsheet renderer	Unknown. Named nowhere in the digest	This document's own check	Blocking question for Stage 1
G1.3 Reporting (manual creation)	Shipped — <i>"users can generate reports in SandGen"</i>	09_shipped-state.md, 2026-08-09	Not a dependency. Named because this module automates what it left manual and must not break it
G1.7 Reporting + Mobile	In progress , reporting half not separately confirmed	09_shipped-state.md, 2026-08-09	Whether this module closes that half is unknown, not assumed. Open question
D0.5 O&M, Billing & Reporting (SandDist)	Slipped. SandDist development not started, expected around September 2026	09_shipped-state.md, 2026-08-09	Not a dependency. It is the reuse case that justifies platform placement
P0.5 APIs	In progress	09_shipped-state.md, 2026-08-09	Nothing here is third-party-facing, so not a dependency. Named so nobody proposes making

Depends on	State	Source	Consequence if not there
			reports available over a public interface in this release

Team context. 09_shipped-state.md records roughly **six engineers and one designer**, with five things simultaneously in progress as of 2026-08-09 and **three of them built but not released**. The brief adds, from the Product Manager on 2026-09-08, that this module displaces the **Sand Generation mobile app**, which was in design — the single designer's queue.

I. Risks in full

Risk	Likelihood / damage	Mitigation
No delivered email route	Near-certain / total for delivery	Confirm before build. Stage 1 proves the renderers without it
A file emailed to the wrong address cannot be recalled	Low / severe	Record before send; the outside-organization privilege; a leadership decision on external recipients
The blocking authority question turns out to be inexpressible	Unknown / high	Private reports can ship without it. Shared and scheduled reports cannot. Answer it first
No scheduler exists	Unknown / high	Blocking question. It changes the size of the work materially
Design capacity	High / moderate	Design Deliverables names what must not be invented, which is where a designer's time actually goes. The module has five design surfaces, two of which (the PDF and the workbook) are easy to forget are design work
Release-and-verification queue	High / moderate	Two dependencies in testing, three unrelated things built-not-released. "Built" is not "available"
Unknown volume	Certain / moderate	Ask before sizing the timer. A thundering herd at 08:00 Monday is the specific failure
Attachment size	Moderate / moderate	The ceiling bar fails visibly. Revisit the download-link non-goal if it bites
Reports diverging from dashboards	Moderate / high	Requirement 7 and Check 16. A report whose numbers differ from the screen destroys trust in both
The tier question left open until a customer asks	Moderate / commercial	Named in Pricing. Whether the waiting account's package includes this is recorded nowhere
"Technical Office" specified as a platform role by accident	Low / moderate	Kept as the client's own wording throughout. Named as an open question, never as a role

J. Rejected alternatives

From the brief's options table, plus the placements rejected in section E.

Option	Why not
Do nothing	The honest baseline. Three teams keep working by hand and the account's ask stays open. Fails the driver
Schedule the default dashboards only — no private or shared reports	<i>Marked as written by the upstream project, not by the Product Manager.</i> The only version that would not wait for per-user accounts or the builders. Not chosen: the Product Manager decided the module ships whole, after both. Worth recording that this document's Stage 1 is close in shape to this option, though as a staging step rather than a smaller release
Report distribution as a shared service every module plugs into	<i>Marked as written by the upstream project, not by the Product Manager.</i> Larger, and the driver is already met without it. It is also the Generative claim this document declines — see The Driver
A smaller first release	Suggested upstream; the Product Manager chose one release on 2026-09-08. Recorded so the choice is visible rather than assumed
The whole module inside `app:sandgen`	See section E
A new reporting microservice	See section E
A second calculation path for reports	See section E

K. Confidence & Flags in full

Where this is solid, and where it is not

Solid. The Product Manager's description of the flow, the three report types, the two output formats, the three PDF content choices, the coverage kinds and the email delivery — all quoted verbatim and reproduced in section D2. The problem statement and the driver evidence, dated and attributed. The roadmap position, re-checked against 09_shipped-state.md rather than taken from the brief. The tier tables, re-read in 01 and 05 rather than accepted from the brief's reading of them. The mechanism inventory and the package layers, from the digest.

Assumption-driven. Every metric target. Every quality bar except the five named as specified. The retention period, the attachment ceiling, the concurrency figure, the repeat-interval set, and three cells of the permission table. The whole verification ladder, which is this document's proposal. The staging, which is this document's proposal. The rollout mechanism, which nobody specified.

Auto-draft disclosure

No interview was run. This PRD was drafted from a confirmed brief plus the knowledge base, per the handoff contract's Step 5.

Which intake batches the brief settled, and which it did not — this document's own assessment, from reading the brief's sections. The brief carries no self-report of this.

Batch	State
1 — Idea, problem, audience	Settled except the product-line and scope-level question, which is never closed by a brief and was decided in Sand Fit
2 — The driver	Evidence supplied, classification decided here. Re-tested against four failure modes; the conclusion matched
3 — Success and scope	Narrowed. No metric exists, correctly. The five exclusions arrived as seeds without reasons or destinations; both were written here, and four more exclusions were added
4 — Users and flows	Narrowed. A seven-step sketch. The privilege model, the permission table, the eleven failure states and the edge cases are this document's
5 — Design	Narrowed only. Surfaces named, no states. The screen inventory, state matrix, interaction notes and open design decisions are all written here — and two surfaces the brief did not name are design work: the PDF layout and the Excel workbook

Batch	State
6 — Data, integration, scale	Narrowed. Routing was matched against the verbatim quotation, which was in ordinary words throughout. No scale figures at all
7 — Verification and rollout	Never closed. Entirely this document's. Twenty-three checks and six stages, every one Architect-proposed
8 — Constraints and context	Prior art settled and re-checked. Constraints narrowed. The approver is [NEEDS INPUT]

Filled by proposal rather than by the Product Manager: the whole acceptance ladder, the staging, the rollout control, the privilege model, the failure states, the screen states, the authority rule, the no-backfill rule, the time-zone requirement, the Excel-from-diagrams rule, and the attachment ceiling.

Architect-proposed items: 14 in the main document and its annex. The one that most needs attention first is **the declared authority in Requirements 19 and 20** — it is the only proposal that might not be expressible in the platform as it stands, and the shared-report half of the module depends on it.

Things that would have been clarifying questions and are now open questions instead: the repeat-interval set, whether a schedule is reusable across reports, whether a Technician uses this module, the retention period, the concurrency figure, and whether reports may leave the client's organization.

Brief provenance

- **Brief ID:** SB-20260908-reporting-module, **draft version v1, contract version v2, content mode PM-provided, Status: Ready for PRD, work type New feature.** The .md file was the one read.
- **Confirmation.** The Handoff Header as received read PM confirmation: Not yet confirmed. **The Product Manager confirmed the brief as sufficient in conversation on 2026-09-08,** and drafting proceeded on that basis. The header row in the brief file is therefore **stale**, and the brief should be reissued at v2 with the row set so the artefact matches the decision. This is recorded because the confirmation gate is the only thing separating a work-in-progress from a specification, and a reader must be able to tell which one they are holding.
- **Predicted content.** Two items were labelled as written by the upstream project rather than by the Product Manager, both in the options table: scheduling default dashboards only, and report distribution as a shared service. **Neither was relied on as evidence.** Both are recorded in section J as rejected alternatives, which is what a hypothesis is for. **The brief's content mode was PM-provided,** so the External driver claim was not a generated one — had the mode been predicted, an External driver would have been a contract violation and the classification would have been rejected.

Every overrule of the brief

What the brief said	What this document concluded	Why
Application scope (declared): Not applicable — platform-level	Partly overruled. Platform for the machinery, application-resident for the setup page and the Generator measure catalogue	"Not applicable" hides a real application surface, and confusing shared machinery with application vocabulary is how a second implementation gets built by accident. Recorded here because a declared scope that was overruled tells the Product Manager where his mental model of the platform needs updating
Dependencies: two, the Admin Module and the builders	Three. A working email route is the third, and it is the one nothing delivers without	Found in PRD-notification-center-v5, which records email as in the pipeline with no date. The brief could not have known this — it does not hold Sand's PRDs
<i>"the dashboards brief ... has never been seen ... should be found and read before the PRD is drafted"</i>	Resolved, not inherited. The dashboards work exists downstream as PRD-customizable-dashboards-v2, dated 2026-08-10, drafted from a brief confirmed 2026-08-09	The brief's similarity check was hedged because the upstream register was last updated 2026-08-11 and never recorded that brief. This document aligned with the PRD instead — including its Requirement 27, <i>"a saved chart or panel can be used to build</i>

What the brief said	What this document concluded	Why
		<i>a report</i> ", and its open question " <i>what does building a report from a saved view actually involve</i> ", which this document answers. The brief register upstream needs the row added
Driver: External	Agreed — but as a conclusion reached here, not a classification inherited	The re-test is described in The Driver. Nothing changed; the checking still had to happen
"Generative is available but not claimed"	Agreed, and declined for a stated reason	Claiming it would license a general delivery framework that the driver does not require
Split decision: one release	Agreed as a release decision, and staged internally in six steps	The brief explicitly permitted this — it kept the parts separate " <i>so they can still be staged inside the release.</i> " Stage 1 in particular is close in shape to the smaller release the Product Manager declined, as a step rather than a shipped subset
Prior art: " <i>looks Plus-shaped</i> "	Refined. The module straddles — pre-defined content is Basic-shaped, custom content is Plus-shaped, and scheduling appears in neither tier list	Re-read 01 and 05 directly. The brief's reading was reasonable and incomplete, and the incompleteness is the commercially interesting part

Seam notes

The seam was clean, and that is worth stating. No package, plane, mechanism or decision name appears anywhere in the brief. No acceptance criterion or verification rung. No price, no rate, no zone, and nothing about Sand's own money. No customer name — the account is identified by pipeline stage throughout, per the standing rule of 2026-08-09. The Data & systems section is three verbatim quotations in the Product Manager's own words with no pre-translation, which is the single most important thing a brief can get right, because the routing in Data & Integration matched against it directly.

Two process notes for upstream, neither a violation. First, the PM confirmation row was left at Not yet confirmed on a brief marked Ready for PRD — a contradiction that stopped the draft until the Product Manager resolved it in conversation. Second, the brief register's missing customizable-dashboards row degraded a similarity check to "hedged" when the neighbor was in fact fully specified downstream. Both are cheap to fix and both cost a round trip here.

Snapshot caveat

Architecture grounding: 08_sand-fit-architecture-digest.md, **generated 2026-08-02** — a compiled snapshot of platform-dev, not a live connection. It records a **live duplicate ADR-031 collision**, and one half of that collision — measure typing and aggregation — is load-bearing in Data & Integration, so **every decision reference in these two documents must be confirmed by filename against the live repository, never by number.** The digest also warns explicitly against inferring decision numbers from its unordered topic list; no number in these documents was taken from that list.

Commercial grounding: 01_company-and-product-overview.md and 05_pricing-packaging-competitive.md, both **September-2025** snapshots. 09_shipped-state.md is human-maintained, **last updated 2026-08-09**, wins over the older roadmap wherever it speaks — and is **behind the Product Manager** on two rows, the Admin Module and the customizable dashboards, both of which he reports as in user acceptance testing as of 2026-09-08. **That file should be updated.**

The live repository and the current price sheet are the final authority, not these documents.

Known source ambiguities touched — none resolved silently

Entry	How it was handled here
A1 — SandGen tier names, Plus versus	Pricing states which row was used (05's SandGen table, upper tier named Advanced,

Entry	How it was handled here
Advanced	glossed " <i>Custom Reports & Billing</i> "), states that nothing in Sand's material says Plus and Advanced are the same tier, and treats the equivalence as assumed and undocumented . Not used as synonyms. The finding that scheduling appears in neither tier list is additional and belongs to Commercial
A7 — two live SandGen role models, four versus eight	Users & Scenarios says outright that the permission table is built on the Admin Module's four , because those are what the software enforces, and that no mapping to the commercial eight is proposed . Beyond that, Requirement 18 writes every action against a privilege , which is the standing guidance for surviving the coming custom-roles release. Three cells are left [NEEDS INPUT] rather than guessed
A4 — the ADR-031 collision	Every decision is cited with the digest's date, the two ADR-031 files are named by title rather than number, and the collision is flagged in both documents
A3 — CMMS parity	Not touched. This document makes no competitive claim

Marker record

Every proposal, assumption and unknown in these two documents is disclosed in one of five ways, and all five appear in both files: **Architect-proposed — confirm** for anything this document proposes rather than knows; [NEEDS INPUT: ...] for a gap left open rather than filled with a plausible figure; [PROPOSED — not measured] for a metric target nobody has measured; "*marked as written by the upstream project*" for the two upstream hypotheses; and **must be confirmed with architecture** on every placement, package, mechanism and decision claim in Sand Fit and Data & Integration.

Nothing was cleaned up because it looked unfinished. Where these documents are unfinished, saying so is the point.